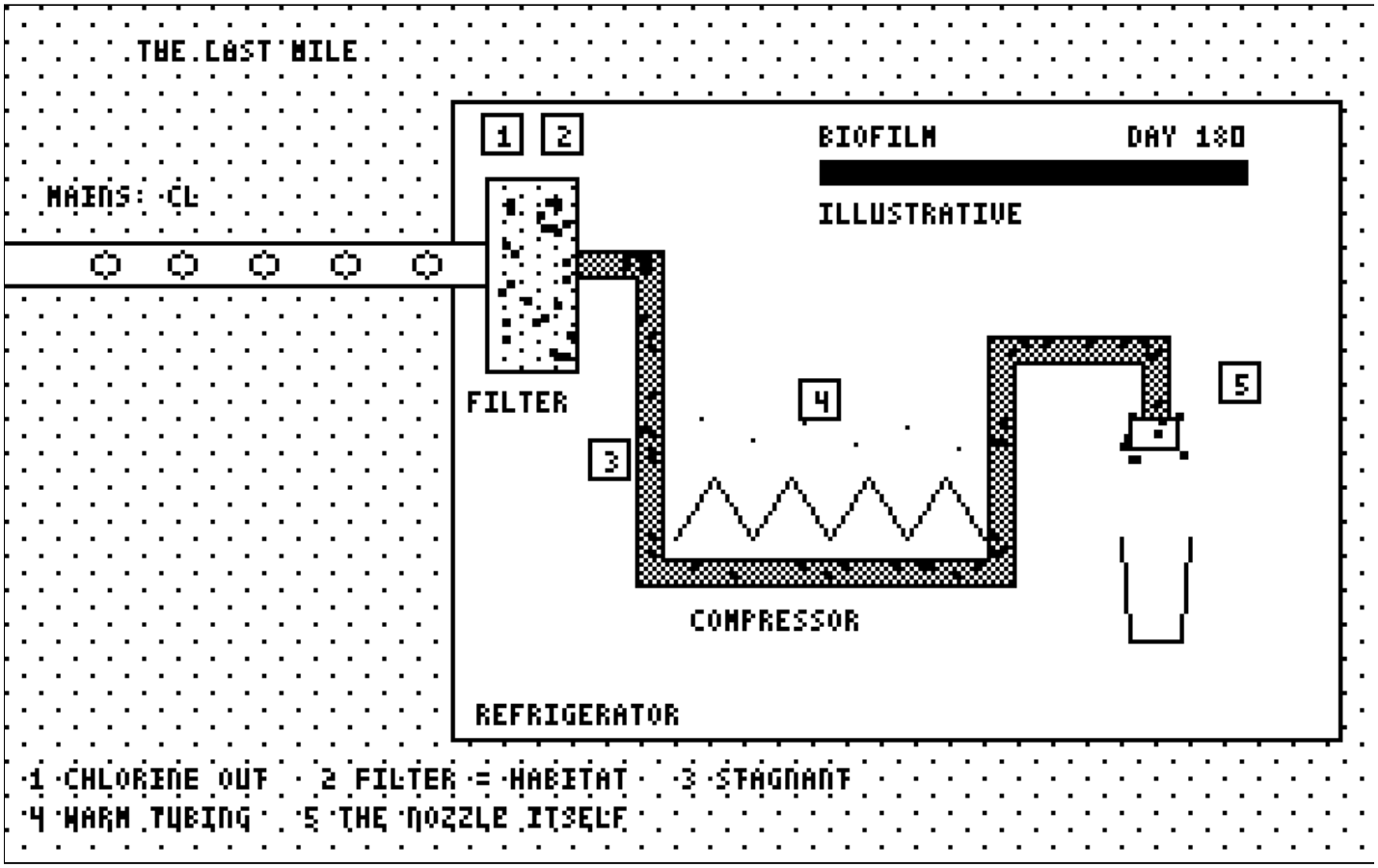
The Hygiene Trap

Scene 3 of 28: The last mile

Last Mile

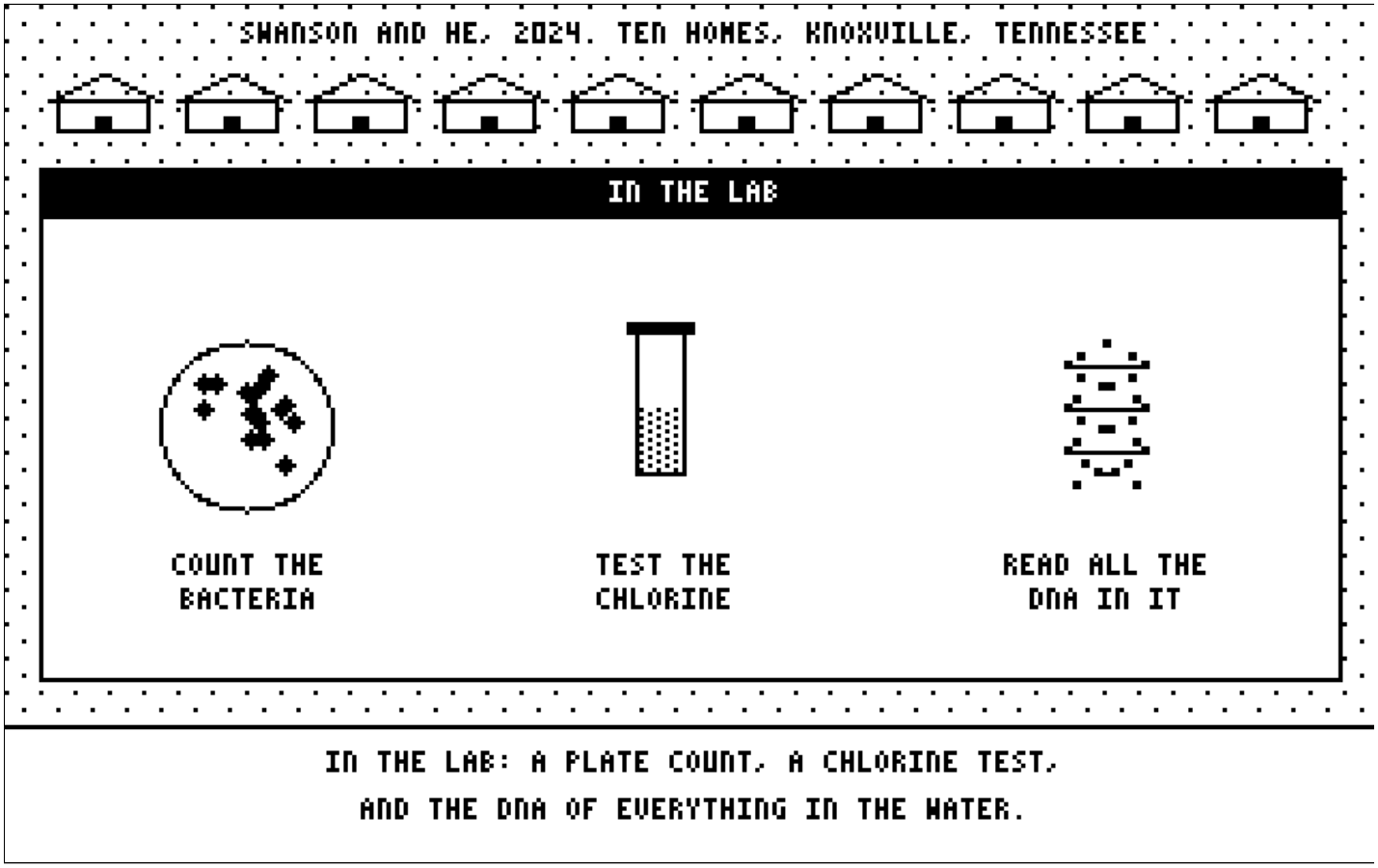


Frame 1. Illustrates line 6: “This part is established. In a 2024 study of…”

What the narrator says

1. Tap water stays safe largely because of a disinfectant, chlorine or chloramine, moving briskly through pressurised pipes.
2. Then it enters the fridge, and five things change.
3. One: the carbon filter takes the disinfectant out, by design. Two: the filter becomes a habitat, because carbon holds organic matter, which is food.
4. Three: the water stops moving, and sits for hours. Four: parts of the tube run beside the compressor, and stay warm.
5. Five: the nozzle, touched by hands and open to kitchen air, picks up whatever is around.
6. This part is established. In a 2024 study of ten homes, fridge water carried thousands of times more bacteria than the tap feeding it, and the chlorine was gone.

The study: ten homes

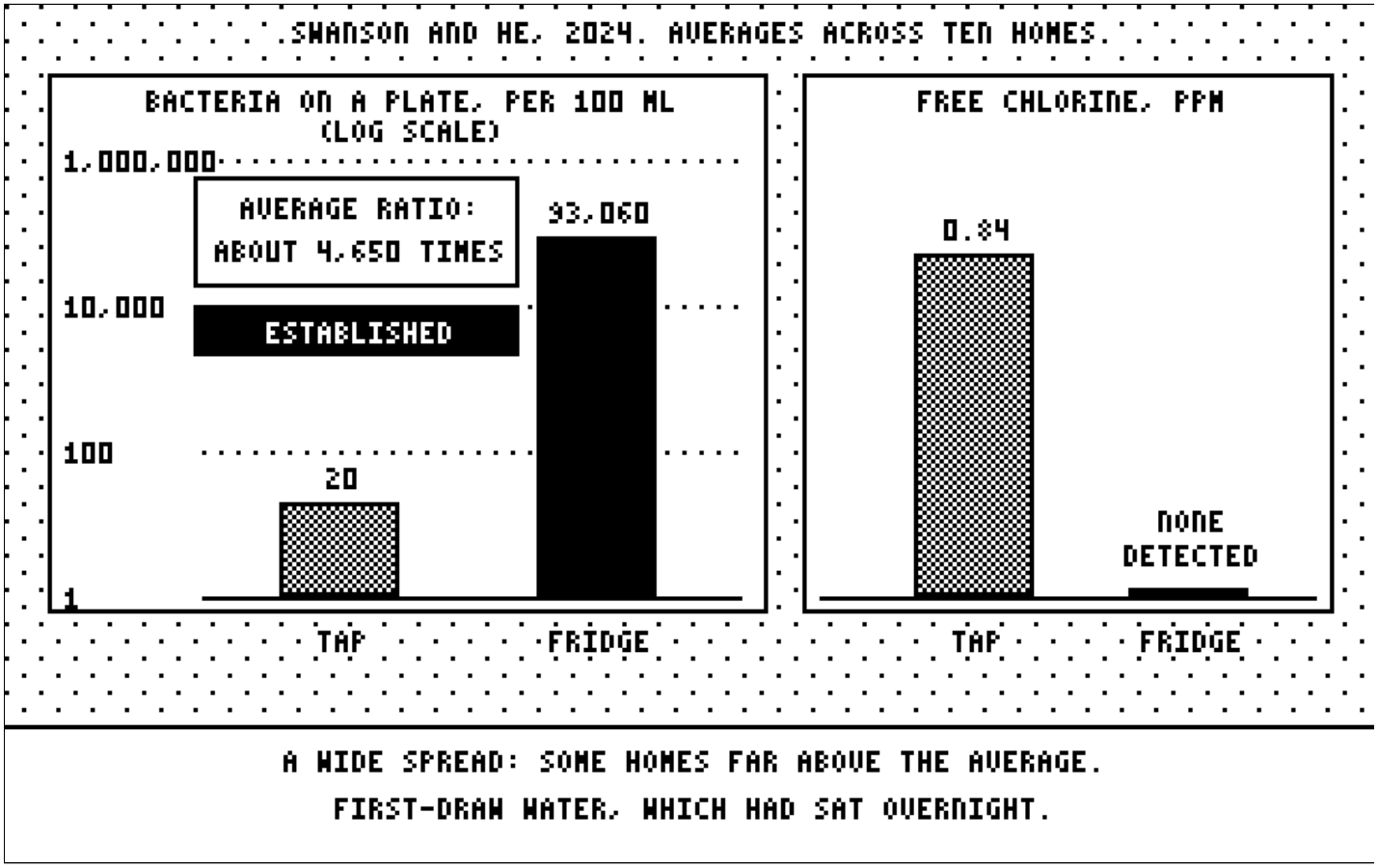


Frame 1. Illustrates line 4: “Back in the lab, they counted the bacteria that…”

What the narrator says

1. Here is the study behind this. Researchers at the University of Tennessee went to ten homes in Knoxville.
2. Each had a fridge with a dispenser and a carbon filter, on a chlorinated public water supply.
3. Early in the morning, before anyone used any water, they filled one litre from the fridge and one from the cold tap of the same home.
4. Back in the lab, they counted the bacteria that grow on a plate, tested the chlorine, and read the DNA of everything in the water.

The result: tap against fridge

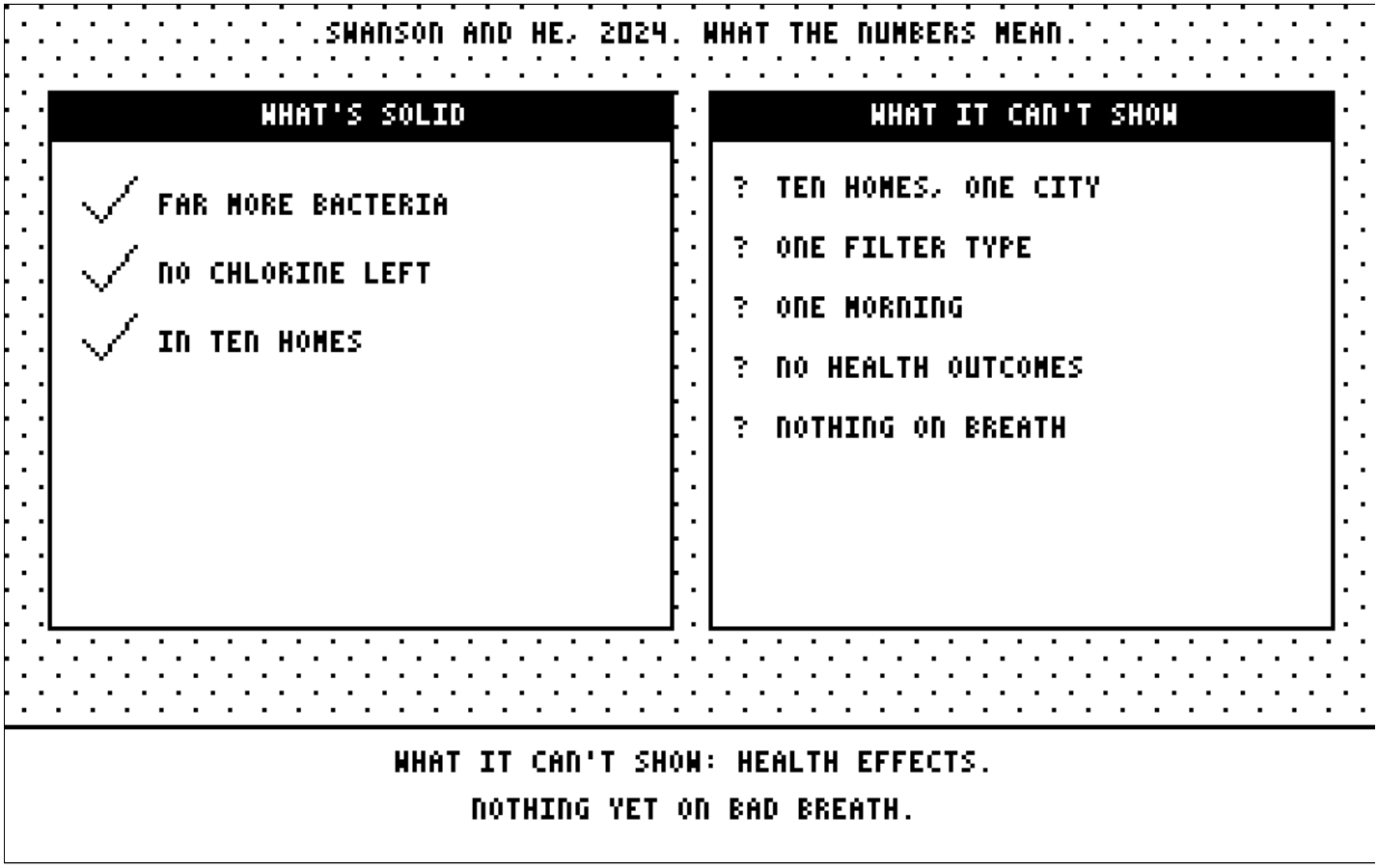


Frame 1. Illustrates line 4: “The spread was wide, so some homes were far…”

What the narrator says

1. The tap water had almost nothing that grows on a plate: about two bacteria in every ten millilitres.
2. The fridge water averaged about nine hundred in every millilitre. That's thousands of times more.
3. And the chlorine was gone. The tap had point eight four parts per million. In the fridge water, none could be detected.
4. The spread was wide, so some homes were far above the average. And these were first-draw samples, water that had sat overnight.

What the numbers mean



Frame 1. Illustrates line 6: “What it can't show: it's ten homes, one city,…”

What the narrator says

1. You may have seen the headline figures: Mycobacterium up eight thousand times, Pseudomonas up two hundred. Here is what they mean.
2. In the tap water, these two groups made up about nine and eleven percent of a very small community.
3. In the fridge water, Mycobacterium was still about seven percent, and Pseudomonas almost none. Their share hadn't risen. The whole community had grown.
4. The authors multiplied each share by the total count, which gives an estimate of the amount. It mixes two methods, so handle the exact numbers with care.
5. What's solid: far more bacteria, and no chlorine, in ten homes.
6. What it can't show: it's ten homes, one city, one filter type and one morning, and nobody's health was measured. Nothing yet on bad breath.

References and notes

1. Swanson CS, He Q (2024). Linkage of in-refrigerator water dispensing systems to elevated exposure to microbial contamination at the microbiome scale. Am J Infect Control 52:857-859. PMID 38460731. doi:10.1016/j.ajic.2024.03.001
2. Zhou Z, Xu L, Zhu L, et al. (2021). Metagenomic analysis of microbiota and antibiotic resistome in household activated carbon drinking water purifiers. Environ Int 148:106394. PMID 33486296. doi:10.1016/j.envint.2021.106394

Notes

- Zhou 2021 was read at the abstract level.
- Raw sequences: NCBI BioProject PRJNA916064 (runs SRR22894290 to SRR22894309).
- The 8,053 and 221 fold changes are the authors' estimate: each genus's share of the sequences multiplied by the total plate count.

Source: The Hygiene Trap, Scene 3 of 25: The last mile. https://fridge-water.ownhealth.bio/fridge-water#s3 (whole site: https://fridge-water.ownhealth.bio/). Printed November 6, 2026.
Every claim carries a label: established, plausible, untested, anecdote or real. This is a hypothesis, not a diagnosis: for persistent bad breath or symptoms, see a dentist and a doctor.
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